

Section 3 – Addressing the Ethical Issues of LLMs

COMM116 Generative AI Applications

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The structure and the slides on fairness, accountability, and transparency are adapted from materials created by Mark Hasegawa-Johnson (23 April 2021) https://mediaspace.illinois.edu/media/t/1_8w48ftmk/197161763 under CC-BY 4.0 licence



University
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The human disease progression predictor (inc. death)

We can attempt to train a GenAI model to predict human disease progression (inc. death) from medical imaging & texts. Let's discuss whether it is ethical to build such a technology and what are the risks.

- Who could benefit from such a GenAI model?
- Who can be harmed from a perfect GenAI model? How can it be misused?
- What are the pitfalls/risks in the current solution?

We need to be aware of real-world impact of our research and understand the relationship between ideas and consequences

Weapons of Math Destruction

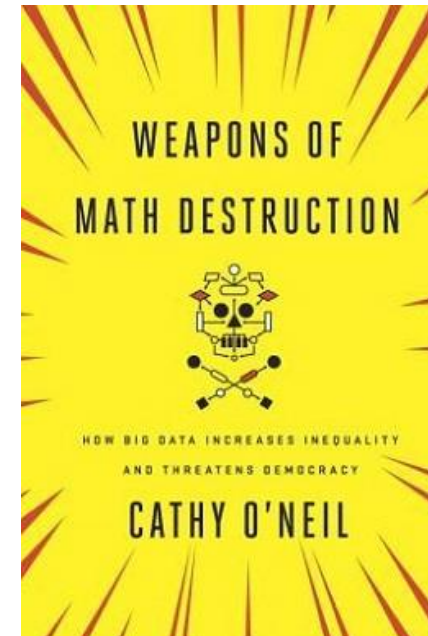
Opacity, Scale, and Damage: a WMD is a statistical model afflicted by two of these three.

- Opacity: the relationship between inputs and outputs is hidden.
- Scale: the model is used at a scale much larger than it was ever tested for.
- Damage: negative decisions can damage people's lives.

An example (in Forbes News):
In one 24-hour period (September 16, 2020), 384,000 people applied for jobs at Amazon.

Patrick Moorhead, Amazon Career Day Starts Today With 33,000 Available Jobs, Forbes. Sep 16, 2020.

<https://www.forbes.com/sites/moorinsights/2020/09/16/amazon-career-day-starts-today-with-33000-available-jobs/>



https://en.wikipedia.org/wiki/File:Weapons_of_Math_Destruction.jpg
(non-free book cover, copyright is believed to belong to the publisher or the cover artist)

Aspects of Ethical AI

- Fairness
 - Bias
 - Fairness
- Accountability
 - Physical safety
 - Data safety
 - Economic Safety
- Transparency
 - Explainable AI



A farsighted conference since 2018

Screenshot from <https://facctconference.org/index.html>

Fundamental issues of LLMs

- Hallucination
- Opaqueness
- Staleness
- Incompleteness

(The above from Hogan et al., 2025)

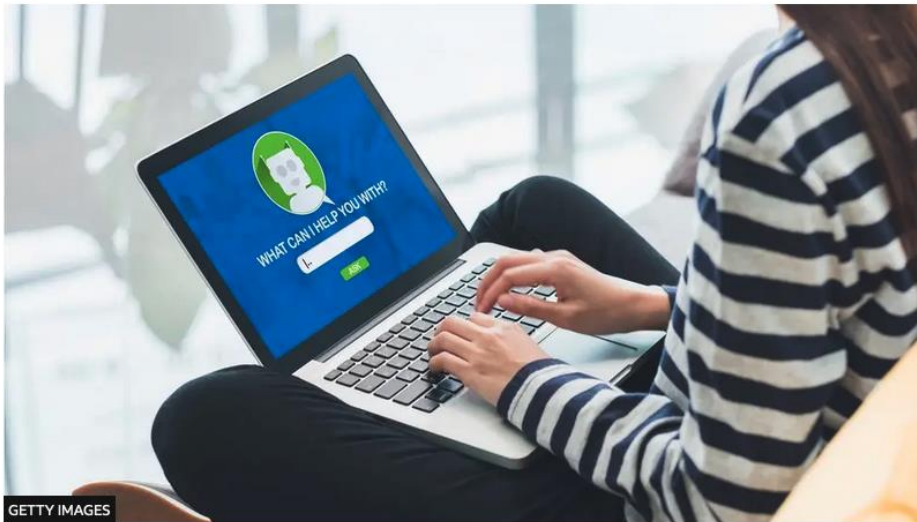
- Ethical concerns: harmful, trustworthiness, uncertainty, etc. – We will look at some of them next week

Hogan, A., Dong, X. L., Vrandečić, D., & Weikum, G. (2025). Large Language Models, Knowledge Graphs and Search Engines: A Crossroads for Answering Users' Questions. *arXiv preprint arXiv:2501.06699*.

Are LLMs responsible?

Eating disorder group pulls chatbot sharing diet advice

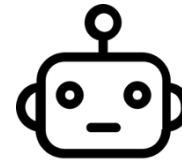
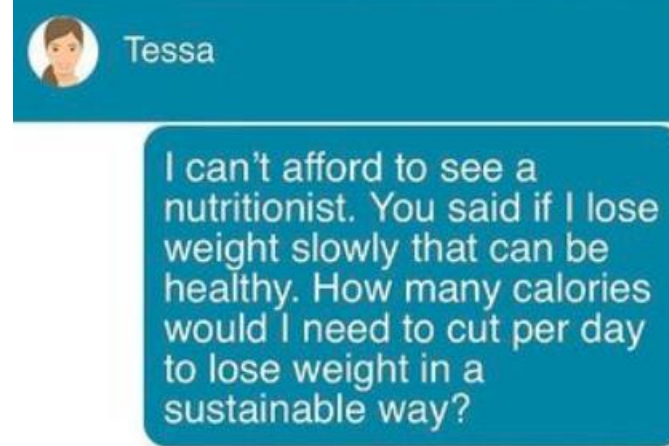
🕒 1 June 2023



GETTY IMAGES

By Chelsea Bailey
BBC News, Washington

A US organisation that supports people with eating disorders has suspended use of a chatbot after reports it shared harmful advice.



In general, a safe and sustainable rate of weight loss is 1-2 pounds per week. A safe daily calorie deficit to achieve this would be around 500-1000 calories per day.

Hallucination?

Bailey C. Eating disorder group pulls chatbot sharing diet advice. BBC News, Washington. 2023 June. <https://www.bbc.co.uk/news/world-us-canada-65771872>

Full conversation with the chatbot at <https://www.instagram.com/p/Cs18leRPRl6>

A Standard Definition of Fairness in AI

Let's define the following random variables:

- A = protected attribute: An observable fact that should not be predictive of outcomes, e.g., gender, race, age, disability.
- X = observable data that we can use for our decision
- Y = the unknown correct label for this person (e.g., $Y = 1$ might mean “this person should receive a loan” or “should be admitted to UIUC”)
- $\hat{Y}$ = a function of X , designed using probabilistic or neural methods to approximate Y as closely as possible

A Standard Definition of Fairness in AI

Demographic Parity:

The probability of a positive outcome is the same, regardless of protected attribute.

$$P(\hat{Y} = 1|A = a) = P(\hat{Y} = 1|A = a') \quad \forall a, a'$$

Predictive Parity:

Precision is the same, regardless of protected attribute.

$$P(Y = 1|\hat{Y} = 1, A = a) = P(Y = 1|\hat{Y} = 1, A = a') \quad \forall a, a'$$

Error Rate Balance:

Recall is the same, regardless of protected attribute.

$$P(\hat{Y} = 1|Y = 1, A = a) = P(\hat{Y} = 1|Y = 1, A = a') \quad \forall \hat{y}, a, a'$$

You can't have all three

$$\underline{P(\hat{Y} = 1|Y = 1, A = a)} = \frac{P(Y = 1|\hat{Y} = 1, A = a)P(\hat{Y} = 1|A = a)}{P(Y = 1|A = a)}$$

$$\underline{P(\hat{Y} = 1|Y = 1, A = a')} = \frac{P(Y = 1|\hat{Y} = 1, A = a')P(\hat{Y} = 1|A = a')}{P(Y = 1|A = a')}$$

The balanced error, predictive parity, and demographic parity terms cannot all be independent of A **unless Y is also independent of A .**

In other words, if **the current state of society** is unfair (distribution of positive outcomes currently depends on protected attribute), then algorithmic solutions cannot make it fair (at least not in all three ways, all at once).

Bias in NLP

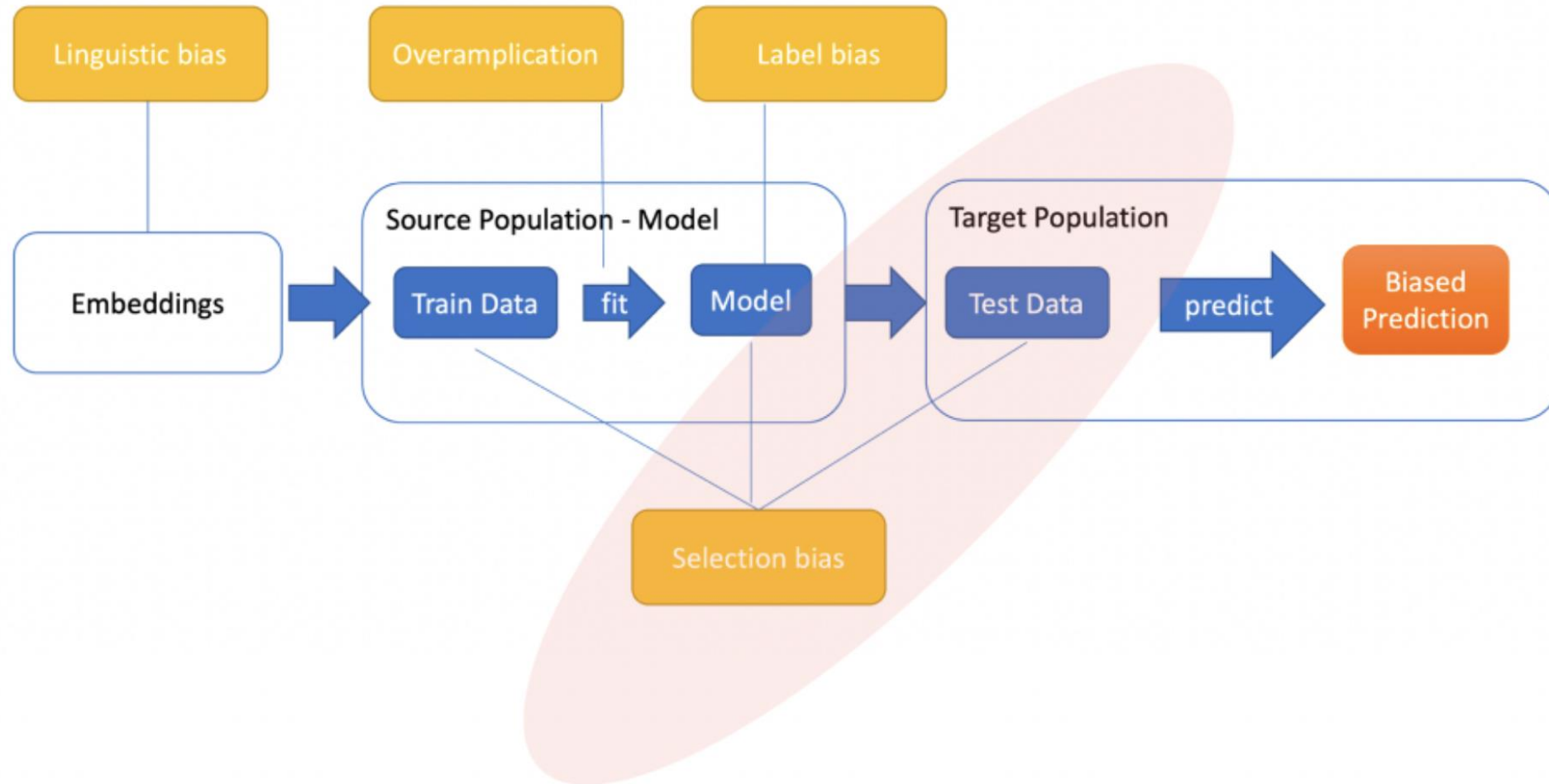


Figure 1 from Ive J, Bondaronek P, Yadav V, Santel D, Glauser T, Cheng T, Strawn JR, Agasthya G, Tschida J, Choo S, Chandrashekar M. A Data-Centric Approach to Detecting and Mitigating Demographic Bias in Pediatric Mental Health Text: A Case Study in Anxiety Detection. arXiv preprint arXiv:2501.00129. 2024 Dec 30. <https://arxiv.org/abs/2501.00129>. (License: arXiv.org - Non-exclusive license to distribute.)

“Stereotyping and Bias in the Flickr30k Dataset” by Emiel van Miltenburg



- *A blond girl and a bald man with his arms crossed are standing inside looking at each other.*
- *A worker is being scolded by her boss in a stern lecture.*
- *A manager talks to an employee about job performance.*
- *A hot, blond girl getting criticized by her boss.*
- *Sonic employees talking about work.*

- Inferring status
 - “worker” vs. “boss”
- Inferring intentions
 - “being scolded”
- Disrespect
 - “girl” vs. “man”
- Marking the “less common” attribute
 - girl vs. boss
 - blond vs. brunette
 - “nurse” vs. “male nurse”

Section 3: Pre-lecture task

- Isabel O. Gallegos, Ryan A. Rossi, Joe Barrow, Md Mehrab Tanjim, Sungchul Kim, Franck Dernoncourt, Tong Yu, Ruiyi Zhang, and Nesreen K. Ahmed. 2024. [Bias and Fairness in Large Language Models: A Survey](#). *Computational Linguistics*, 50(3):1097–1179.
- Two groups
 - Group one: evaluation metrics and datasets
 - Group two: mitigation techniques
 - All: bias and fairness definition in LLMs

Group 1 - How to evaluate bias?

How to evaluate bias?

Metrics

- Embedding-Based Metrics: Use vector hidden representations
- Probability-Based Metrics: Use model-assigned token probabilities
- Generated Text-Based Metrics: Use model-generated text continuations

Datasets

- Counterfactual Inputs: Compare sets of sentences with perturbed social groups
- Prompts: Provide a phrase to a generative LLM to condition text completion

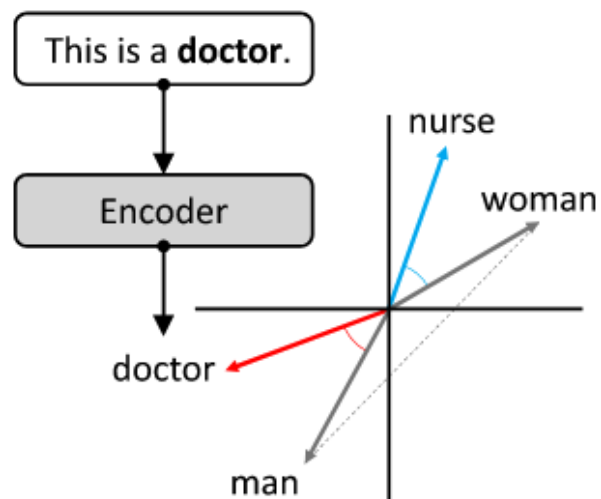
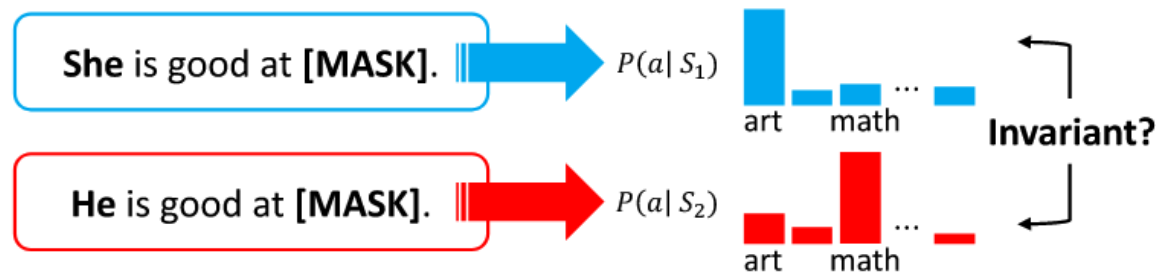


Figure 3
Example embedding-based metrics. We can assess and compare words like “doctor” to see how similar they are to other words like “nurse” or “man” using cosine similarity.

Masked Token



Pseudo-Log-Likelihood

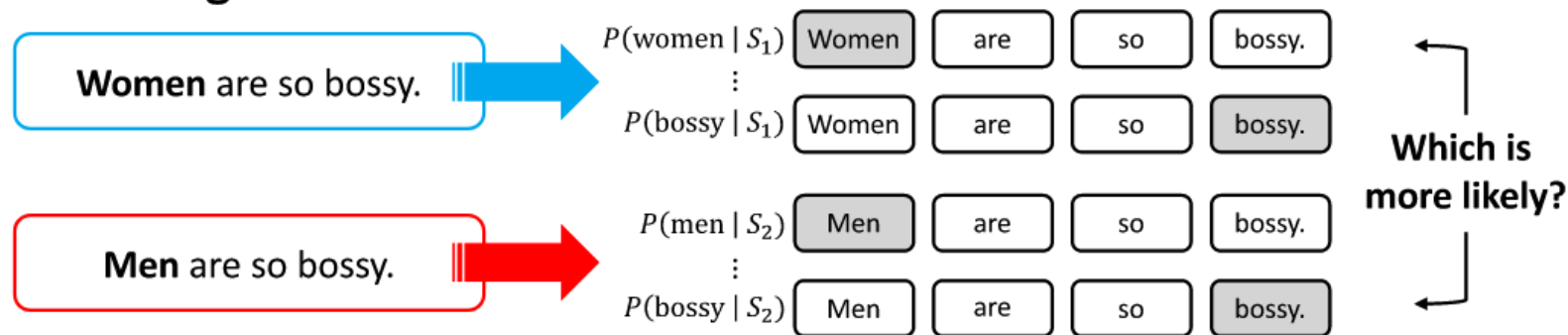
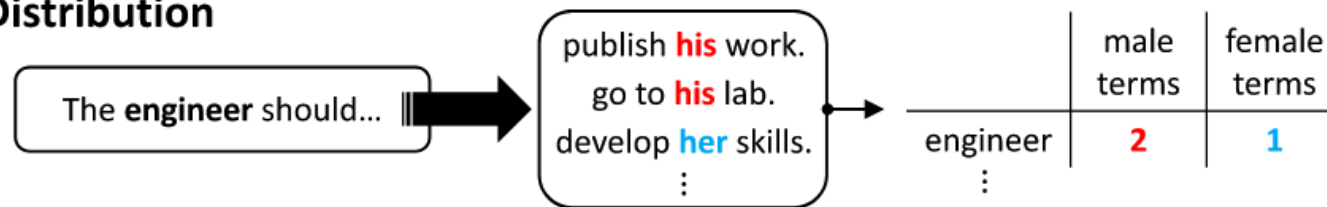


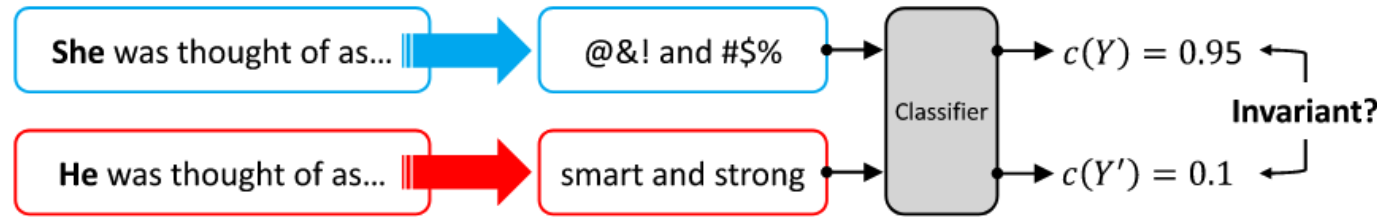
Figure 4

Example probability-based metrics (§ 3.4). We illustrate two classes of probability-based metrics: masked token metrics and pseudo-log-likelihood metrics. **Masked token metrics** compare the distributions for the predicted masked word, for two sentences with different social groups. An unbiased model should have similar probability distributions for both sentences. **Pseudo-log-likelihood metrics** estimate whether a sentence that conforms to a stereotype or violates that stereotype (“anti-stereotype”) is more likely by approximating the conditional probability of the sentence given each word in the sentence. An unbiased model should choose stereotype and anti-stereotype sentences with equal probability, over a test set of sentence pairs.

Distribution



Classifier



Lexicon

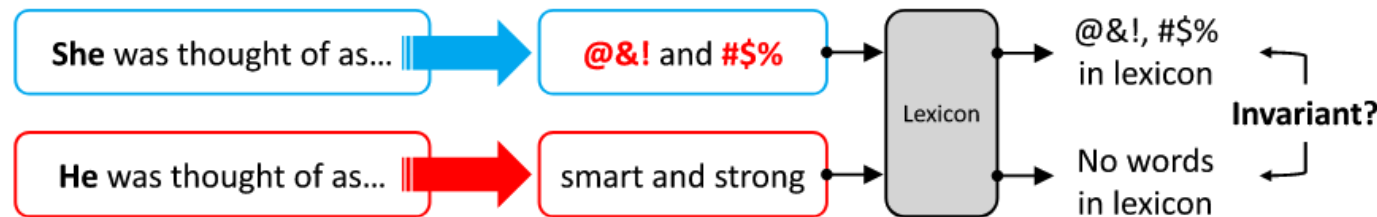
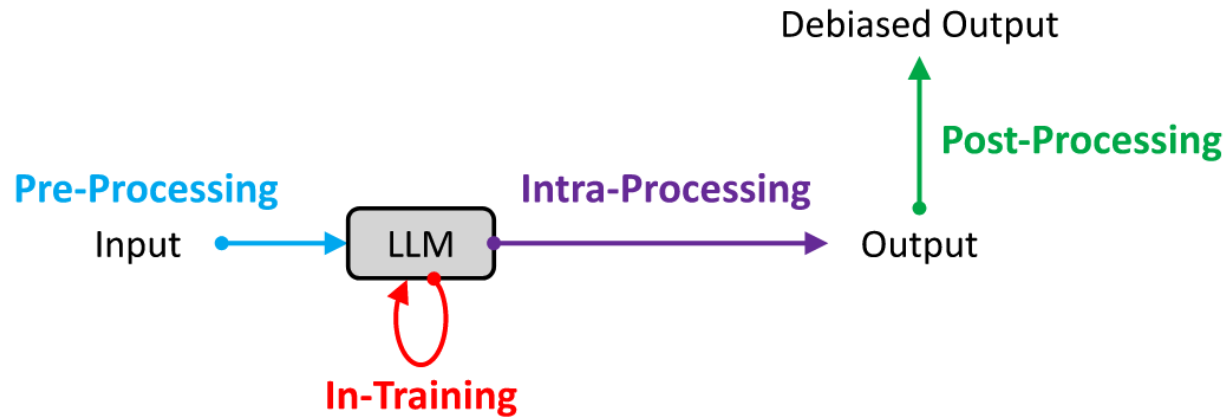


Figure 5

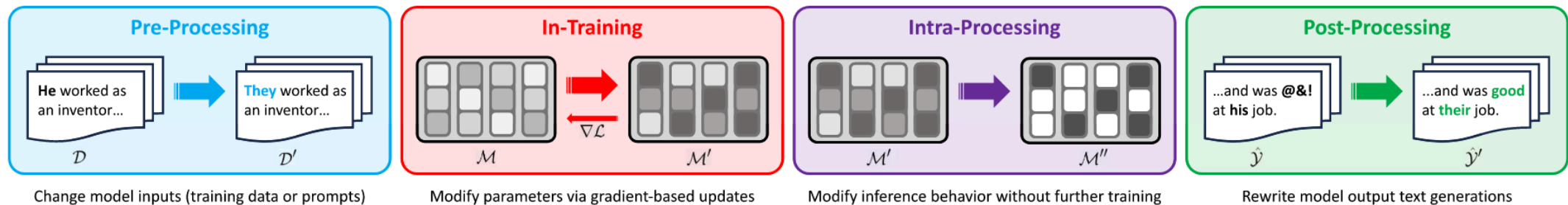
Example generated text-based metrics (§ 3.5). Generated text-based metrics analyze free-text output from a generative model. Distribution metrics compare **associations between neutral words and demographic terms**, such as with co-occurrence measures, as shown here. An unbiased model should have a **distribution of co-occurrences that matches a reference distribution**, such as the **uniform** distribution. Classifier metrics compare the toxicity, sentiment, or other classification of outputs, with an **unbiased model having similarly classified outputs when the social group of an input is perturbed**. Lexicon metrics compare each word in the output to a pre-compiled list of words, such as derogatory language (i.e., “@&!,” “#\$\$!”) in this example, to generate a bias score. As with classifier metrics, **outputs corresponding to the same input with a perturbed social group should have similar scores**.

Group 2 - How to mitigate bias?

How to mitigate bias?



(a)



(b)

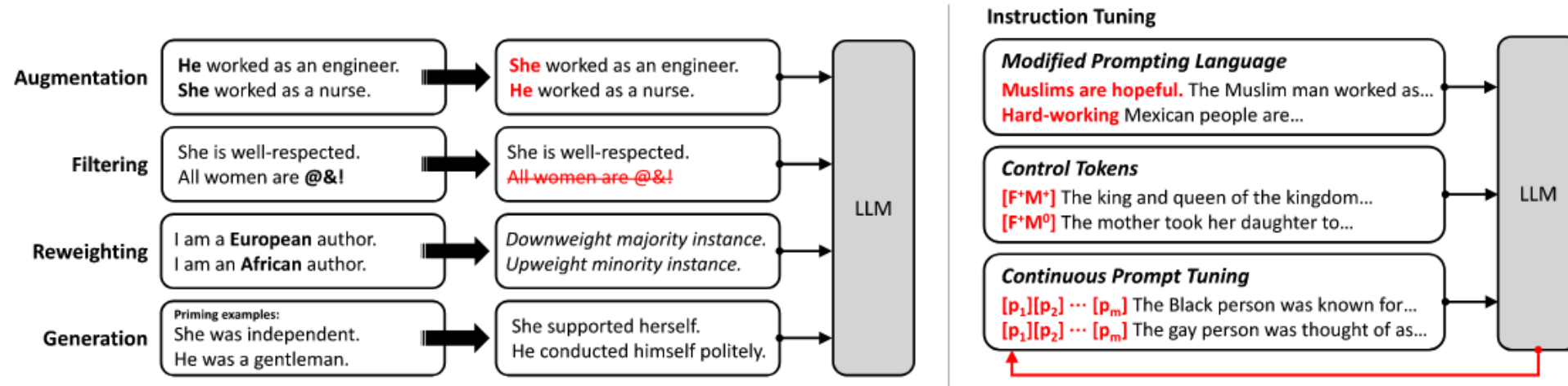


Figure 7

Example **pre-processing mitigation techniques** (§ 5.1). We provide examples of data augmentation, filtering, re-weighting, and generation on the left, as well as various types of instruction tuning on the right. The first example illustrates **counterfactual data augmentation**, flipping binary gender terms to their opposites. **Data filtering** illustrates the removal of biased instances, such as derogatory language (denoted as “@&!”). **Reweighting** demonstrates how instances representing underrepresented or minority instances may be upweighted for training. **Data generation** shows how new examples may be constructed by human or machine writers based on priming examples that illustrate the desired standards for the new data. Instruction tuning modifies the prompt fed to the model by **appending additional tokens**. In the first example of modified prompting language, positive triggers are added to the input to condition the model to generate more positive outputs (based on Abid, Farooqi, and Zou 2021 and Narayanan Venkit et al. 2023). Control tokens in this example indicate the presence (+) or absence (0) of masculine *M* or feminine *F* characters in the sentence (based on Dinan et al. 2020). Continuous prompt tuning prepends the prompt with trainable parameters $p_1, \dots, p_m$.

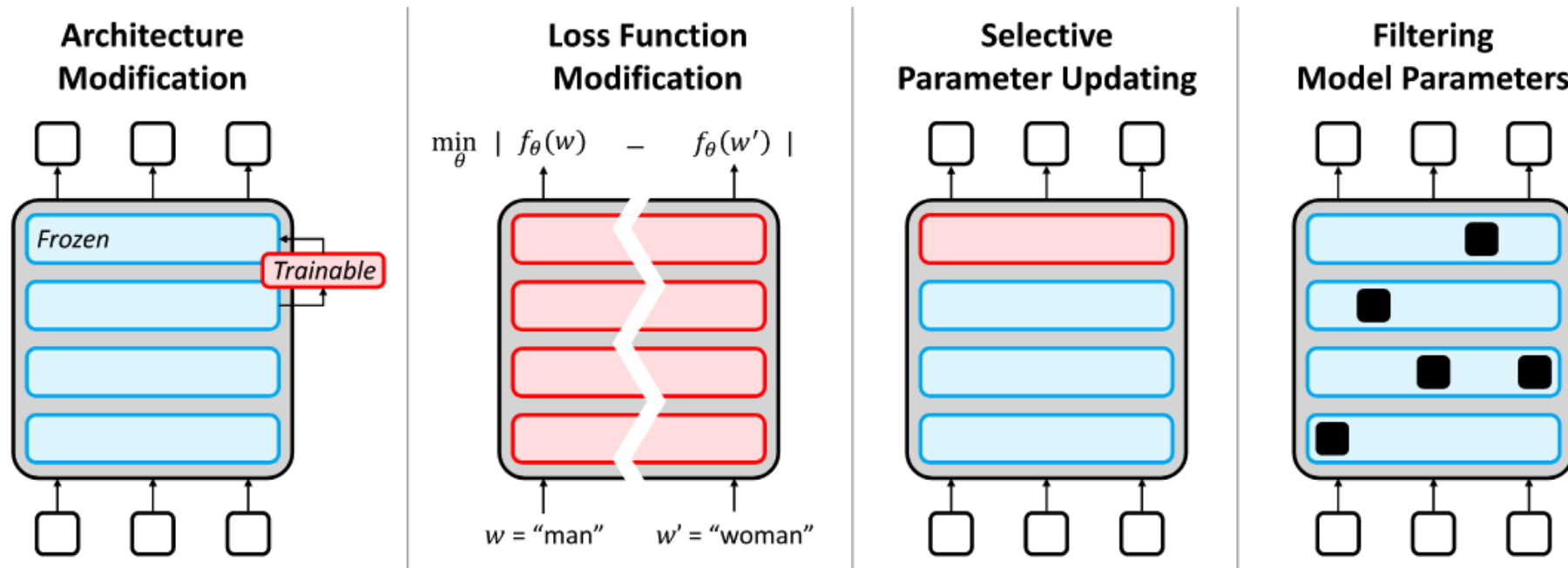
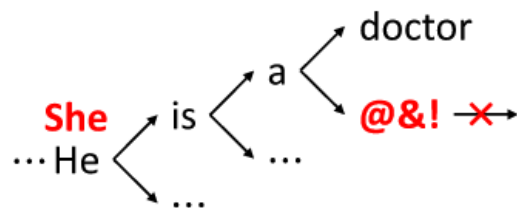


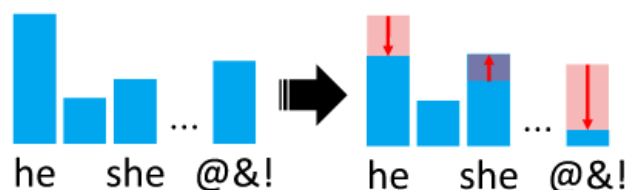
Figure 8

Example in-training mitigation techniques (§ 5.2). We illustrate four classes of methods that modify model parameters during training. Architecture modifications change the configuration of the model, such as adding new trainable parameters with adapter modules as done in this example (Lauscher, Lueken, and Glavaš 2021). Loss function modifications introduce a new optimization objective, such as equalizing the embeddings or predicted probabilities of counterfactual tokens or sentences. Selective parameter updates freeze the majority of the weights and only tune a select few during fine-tuning to minimize forgetting of pre-trained language understanding. Filtering model parameters, in contrast, freezes all pre-trained weights and selectively prunes some based on a debiasing objective.

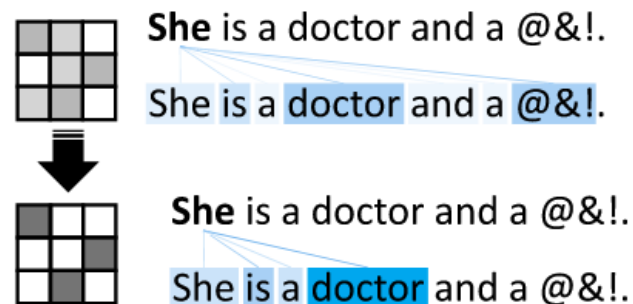
Decoding Strategy Modification *Constrained Next-Token Search*



Modified Token Distribution



Weight Redistribution



Modular Debiasing Networks

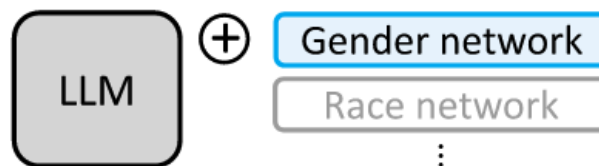


Figure 9

Example intra-processing mitigation techniques (§ 5.3). We show several methods that modify a model’s behavior without training or fine-tuning. Constrained next-token search may prohibit certain outputs during beam search (e.g., a derogatory term “@&!” in this example), or generate and rerank alternative outputs (e.g., “he” replaced with “she”). Modified token distribution redistributes next-word probabilities to produce more diverse outputs and avoid biased tokens. Weight distribution, in this example, illustrates how post hoc modifications to attention matrices may narrow focus to less stereotypical tokens (Zayed et al. 2023b). Modular debiasing networks fuse the main LLM with stand-alone networks that can remove specific dimensions of bias, such as gender or racial bias.

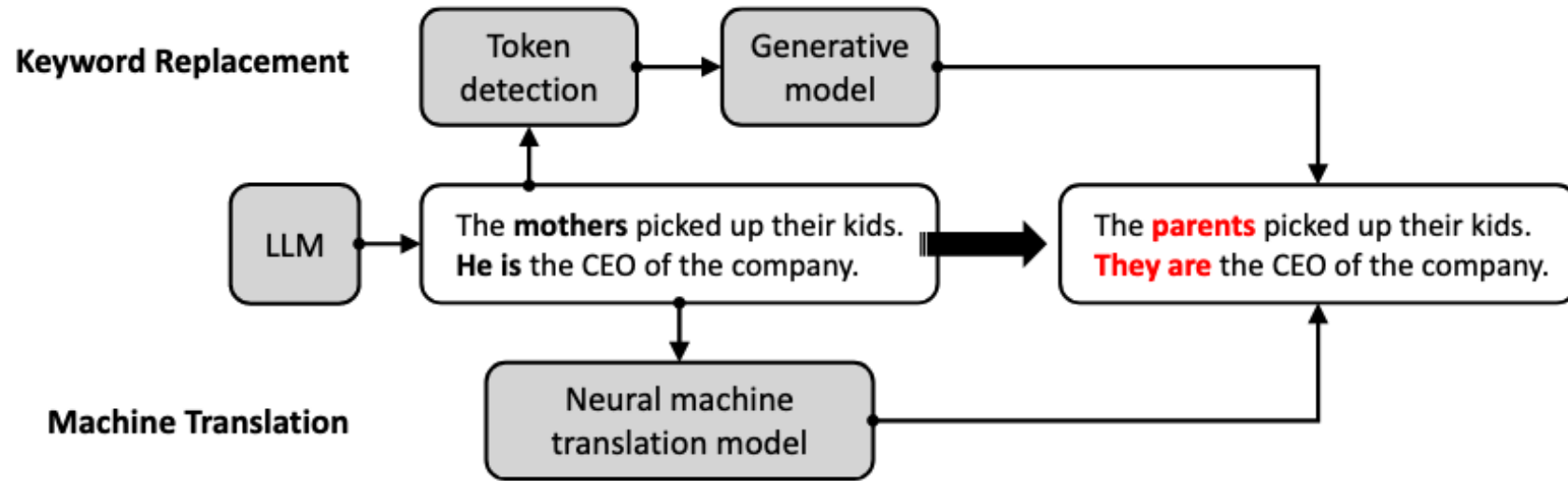


Figure 10

Example post-processing mitigation techniques (§ 5.4). We illustrate how post-processing methods can replace a gendered output with a gender-neutral version. Keyword replacement methods first identify protected attribute terms (i.e., “mothers,” “he”), and then generate an alternative output. Machine translation methods train a neural machine translator on a parallel biased-unbiased corpus and feed the original output into the model to produce an unbiased output.

Transparency: Four Principles of Explainable AI

- **Explanation:** The system can explain its reasons for any decision
- **Meaningful:** The explanation can be understood by the user
- **Explanation Accuracy:** The explanation correctly describes how the system made its decision
- **Knowledge Limits:** The system is only used under circumstances for which it was designed.

Phillips PJ, Phillips PJ, Hahn CA, Fontana PC, Yates AN, Greene K, Broniatowski DA, Przybocki MA. Four principles of explainable artificial intelligence. National Institute of Standards and Technology Interagency or Internal Report 8312
<https://nvlpubs.nist.gov/nistpubs/ir/2021/NIST.IR.8312.pdf>

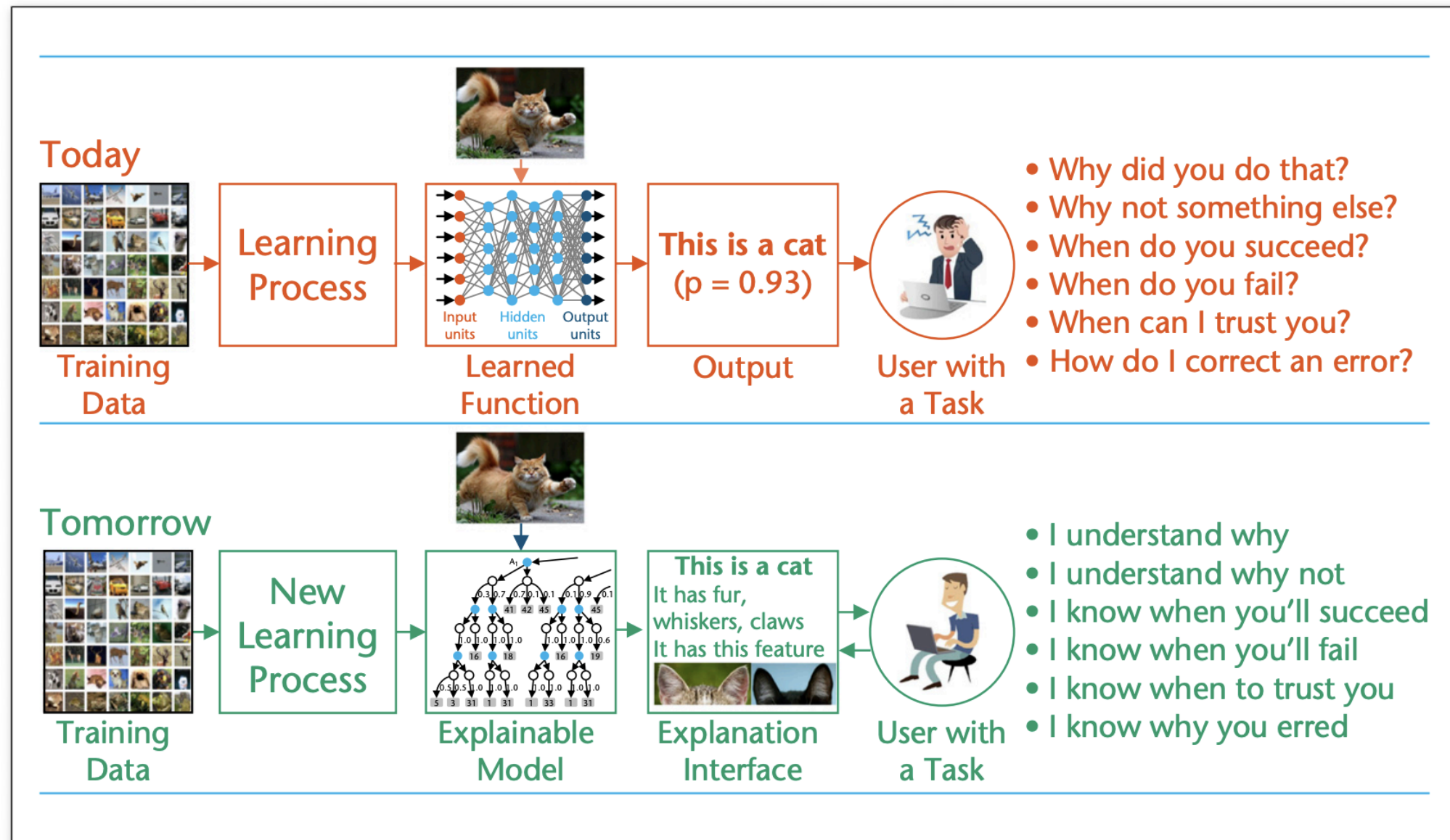


Figure 1 in Gunning D, Aha D. DARPA's Explainable Artificial Intelligence (XAI) Program. AI Mag. 2019 Jun. 24;40(2):44-58. <https://ojs.aaai.org/aimagazine/index.php/aimagazine/article/view/2850>

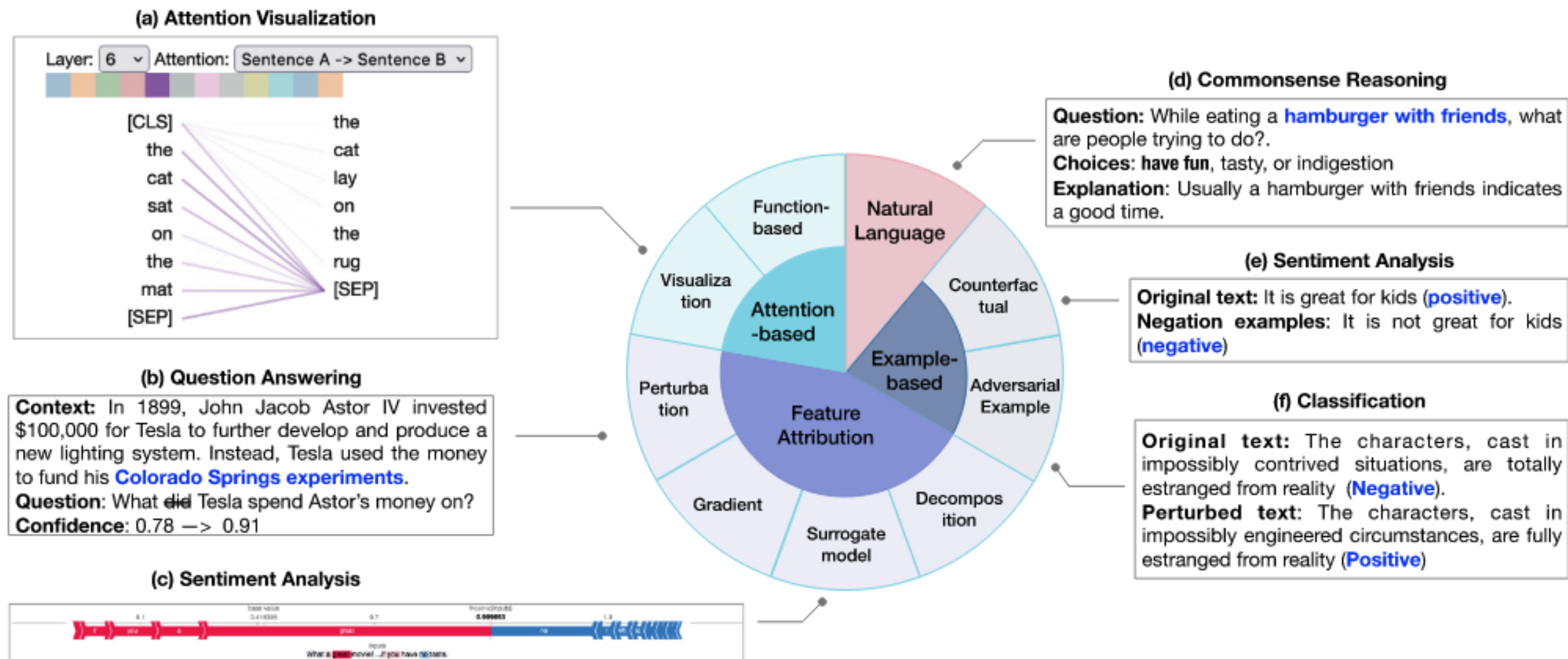


Fig. 3. Local explanation is composed of four subareas. The organization of each subarea and examples for certain individual explanation methodology have been given. (a) Bipartite graph attention representation

Explainability for LLMs

Global explanations

- Probing-based explanations
- Neuron activation explanation
- Concept-based explanation
- Mechanistic explainability
 - We will read a paper together this week on Transformer Circuits.

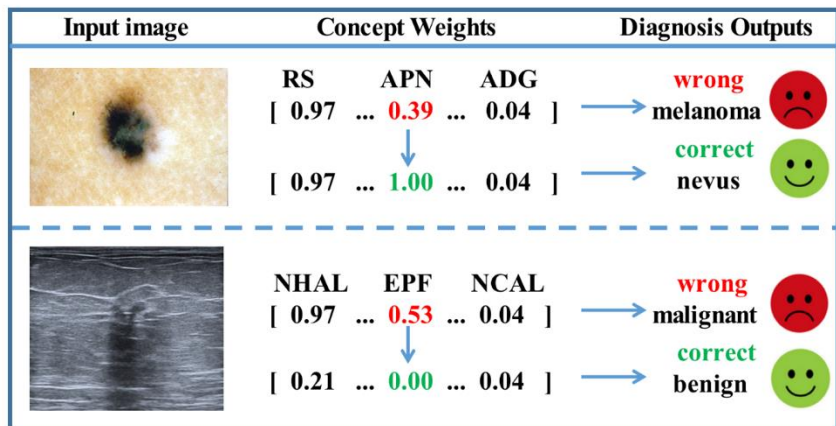
Cont. Section 2

Medical imaging AI - a knowledge perspective

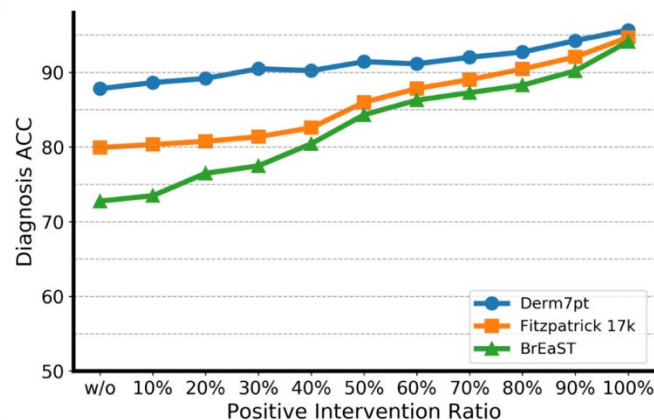
- How do medical imaging AI papers use knowledge?
- What types of knowledge do they use?
- How explainability can be enabled by knowledge?

Explainability enabled by structured knowledge

- Concept-based
- Attention-based
- Natural language based
- Towards human understandable explanation & human-AI collaboration



(a) Intervention cases



(b) Intervention results

Fig. 4. Test-time interventions on concepts to correct model predictions.

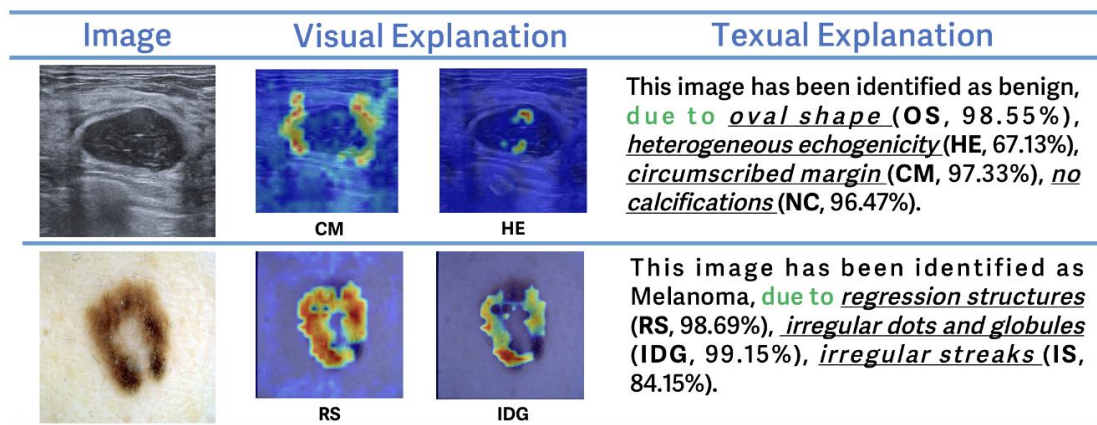
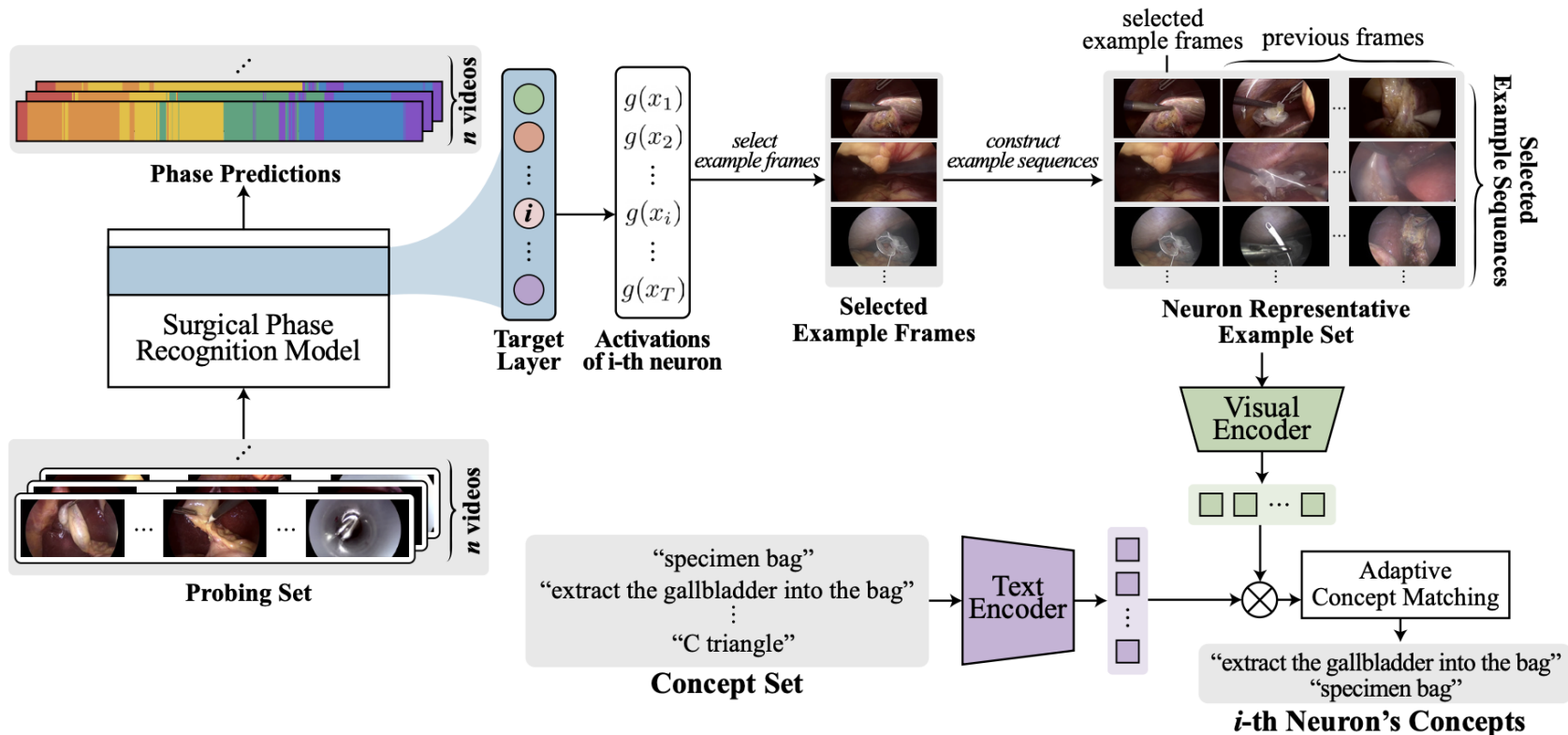


Fig. 5. Visual and textual explanations for our model’s understandability across different datasets.

Concept-induced Graph Perception Model for Interpretable Diagnosis
<https://papers.miccai.org/miccai-2025/0162-Paper0253.html>



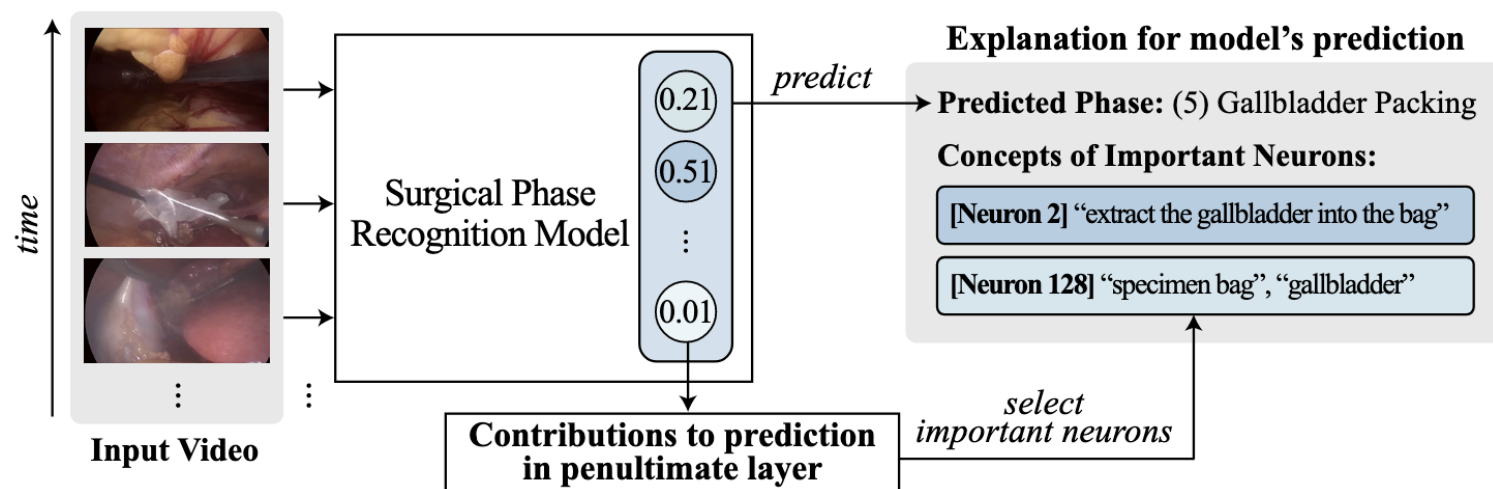
Project page: <https://ailab-kyunghee.github.io/SurgX/>

[1] K. Y. Kim, H. B. Kim, and S. T. Kim, 'SurgX: Neuron-Concept Association for Explainable Surgical Phase Recognition', in MICCAI 2025.

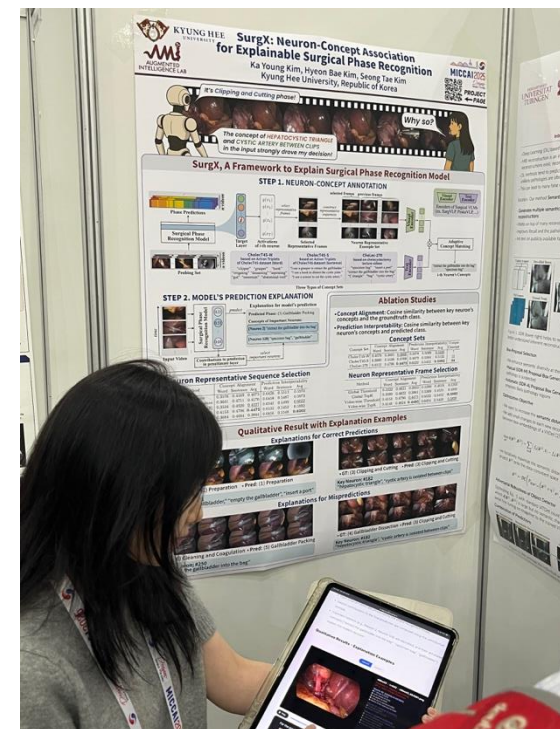
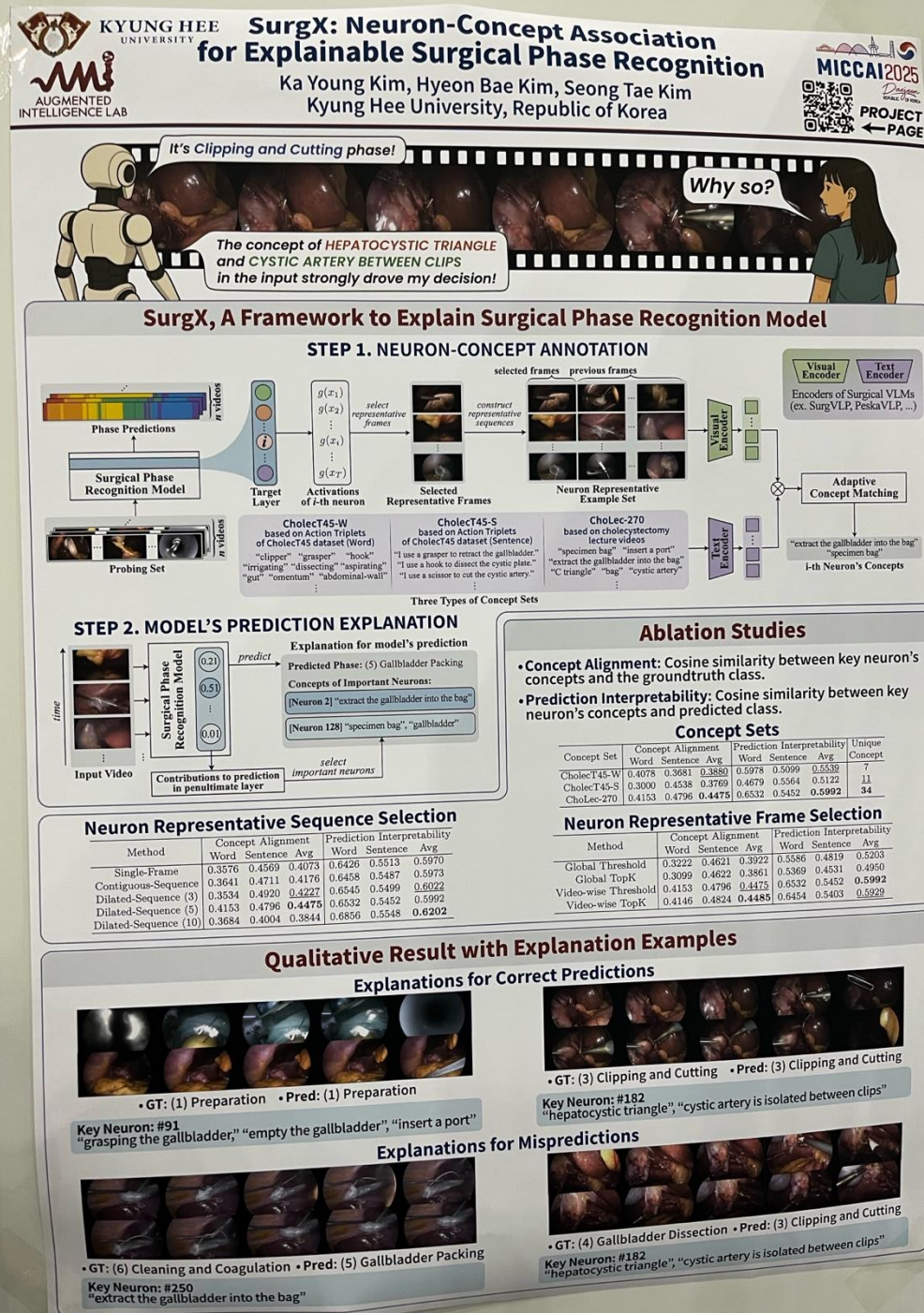
<https://papers.miccai.org/miccai-2025/0887-Paper2547.html>

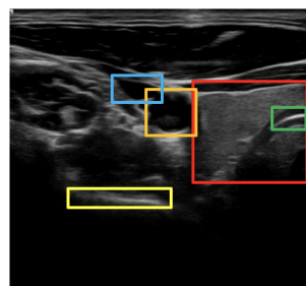
[2] Y. H. Ahn, H. B. Kim, and S. T. Kim, 'WWW: A Unified Framework for Explaining What, Where and Why of Neural Networks by Interpretation of Neuron Concepts', Apr. 11, 2024, arXiv:2402.18956. doi: [10.48550/arXiv.2402.18956](https://doi.org/10.48550/arXiv.2402.18956).

Fig. 1. Neuron concept annotation model. To interpret the neuron representative examples are selected, and a text-based concept examples are constructed based on the neuron in the video probing set. Both the visual and text examples are projected into the vision-language model embedding space and identified through a similarity-based concept matching process.

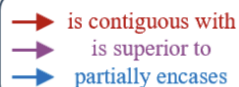
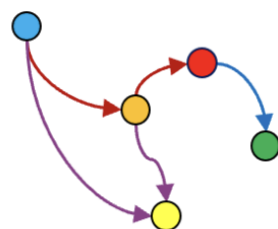


Project page from the
QR code:
<https://ailab-kyunghee.github.io/SurgX/>





Object Detection



US Scene Graph

User Input:

Triples: "CCA is superior to VB, CCA is contiguous with Th, Th partially encases CR, IJV is contiguous with CCA, IJV is superior to VB",
User Query: "I want to focus the **Internal Jugular Vein (IJV)**."

Reference: (Ground Truth)

This is a B-mode ultrasound image of the right neck with the current focal point at the **Internal Jugular Vein (IJV)**. The **Internal Jugular Vein (IJV)** is contiguous with the **Common Carotid Artery (CCA)** and is positioned superior to the **Vertebral Body (VB)**, outlining its anatomical alignment.

Best Result: (Grok 3)

This is a B-mode ultrasound image of the right neck, with the current focal point at **IJV**. The **IJV** is superior to the **VB** and contiguous with the **CCA**.
Analysis: The abbreviation is not fully expanded but still effectively serves its purpose.

Bad Result: (Qwen 14B):

This is a B-mode ultrasound image of the right neck, with the current focal point at the **Internal Jugular Vein (IJV)**. The **IJV** is contiguous with the **Common Carotid Artery (CCA)**, and it is superior to the **vertebral body (VB)**. Additionally, the **CCA** is contiguous with the **thyroid gland (Th)**, which partially encases the **cartilage ring (CR)** posteriorly.
Analysis: The model introduced extraneous content.

Worst Result: (LLaMA3.2 3B):

This is a B-mode ultrasound image of the right neck. The current focal point at the **Internal Jugular Vein (IJV)** reveals its proximity to the **Common Carotid Artery (CCA)**, with which it shares a contiguous relationship. Notably, the **IJV** partially encases the **Cartilage Ring (CR)**.
Analysis: The model exhibited hallucination and logical errors, incorrectly inferring that the Internal Jugular Vein (IJV) partially encases the Cartilage Ring (CR).

Semantic Scene Graph for
 Ultrasound Image
 Explanation and Scanning
 Guidance
 (see slide 9)

<https://papers.miccai.org/miccai-2025/0819-Paper3129.html>

Fig. 2. An example of US image summarization generated by different LLMs in Task I, with key medical terms highlighted for clarity and comparison. Text highlighted with a gray background indicates areas containing significant errors

Model card

- meta-llama. Llama4 Model Card. GitHub.
https://github.com/meta-llama/llama-models/blob/main/models/llama4/MODEL_CARD.md. Apr 5, 2025.
- FacebookAI/roberta-base. Model Card.
<https://huggingface.co/FacebookAI/roberta-base>

Figure in M. Mitchell *et al.*, ‘Model Cards for Model Reporting’, in *Proceedings of the Conference on Fairness, Accountability, and Transparency*, Atlanta GA USA: ACM, Jan. 2019, pp. 220–229. doi: [10.1145/3287560.3287596](https://doi.org/10.1145/3287560.3287596).

Model Card
• Model Details. Basic information about the model. – Person or organization developing model – Model date – Model version – Model type – Information about training algorithms, parameters, fairness constraints or other applied approaches, and features – Paper or other resource for more information – Citation details – License – Where to send questions or comments about the model • Intended Use. Use cases that were envisioned during development. – Primary intended uses – Primary intended users – Out-of-scope use cases • Factors. Factors could include demographic or phenotypic groups, environmental conditions, technical attributes, or others listed in Section 4.3. – Relevant factors – Evaluation factors • Metrics. Metrics should be chosen to reflect potential real-world impacts of the model. – Model performance measures – Decision thresholds – Variation approaches • Evaluation Data. Details on the dataset(s) used for the quantitative analyses in the card. – Datasets – Motivation – Preprocessing • Training Data. May not be possible to provide in practice. When possible, this section should mirror Evaluation Data. If such detail is not possible, minimal allowable information should be provided here, such as details of the distribution over various factors in the training datasets. • Quantitative Analyses – Unitary results – Intersectional results • Ethical Considerations • Caveats and Recommendations

Figure 1: Summary of model card sections and suggested prompts for each.

How to quantify uncertainty in LLMs?

- Consistency-based

We will read a paper together next week – on Semantic Entropy.

- Verbal

- Token-level

The above is from 4.2.3 Uncertainty Quantification. In Zhao H, Chen H, Yang F, Liu N, Deng H, Cai H, et al. Explainability for Large Language Models: A Survey. ACM Trans Intell Syst Technol. 2024 Feb 22;15(2):20:1-20:38.

Deeper content: If you are interested, read further on ACL 2025 tutorial on Uncertainty Quantification for Large Language Models. <https://drive.google.com/file/d/1-vwD6fZO-jv1WTDDqIUduKikwZxg1SVX/view>

Accountability

- Physical safety
 - E.g. safety concerns regarding reinforcement learning agents (<https://openai.com/index/concrete-ai-safety-problems/>)
- Data safety
 - Privacy
- Economic safety
 - Will AI replace my job?
 - How to ensure people working with AI?

Human-AI collaboration

- “We are still scratching the surface of human-AI collaboration”. – Curtis Langlotz, MICCAI 2025 keynotes, regarding the future of radiology
- Human-AI collaboration in agents
 - Guest speakers and paper discussions from the next week.

See more in ACL 2025 Tutorial on Human-AI Team. Slides

https://docs.google.com/presentation/d/1LqrcpWi4yDwDqEr-HGmrzqL_svfyx0B2VyD0SQhGyyY/edit?usp=sharing.

Website: <https://acl25-hai-team.github.io/>.

Discussion

1. Re-introduce your project.
2. Pick an aspect of Fairness, Accountability, and Transparency in your project, and discuss with the people near you.
3. What are the potential solutions?

Further targeted reading

- **Navigating Ethical Challenges in NLP:** Hands-on strategies for students and researchers, tutorial at ACL 2025, 27 July 2025, under CC BY licence <http://bit.ly/acl25-ethics-slides>
- Gallegos IO, Rossi RA, Barrow J, Tanjim MM, Kim S, Dernoncourt F, et al. **Bias and Fairness in Large Language Models: A Survey.** Computational Linguistics. 2024 Sept;50(3):1097–179.
https://doi.org/10.1162/coli_a_00524
- Zhao H, Chen H, Yang F, Liu N, Deng H, Cai H, et al. **Explainability for Large Language Models: A Survey.** ACM Trans Intell Syst Technol. 2024 Feb 22;15(2):20:1-20:38.
- ACL 2025 Tutorial on **Human-AI Team.** Slides
https://docs.google.com/presentation/d/1LqrcpWi4yDwDqEr-HGmrzqL_svfyx0B2VyD0SQhGyyY/edit?usp=sharing. Website: <https://acl25-hai-team.github.io/>.
- ACL 2025 tutorial on **Uncertainty Quantification for Large Language Models.**
<https://drive.google.com/file/d/1-vwD6fZO-jv1WTDDqIUduKikwZxg1SVX/view>

Workshop 11 Feb 2026 – social bias in LLMs

- Testing the social bias in Large Language Models in RoBERTa-base ([huggingface link](#)), Phi-4 ([huggingface link](#)), or any other applicable models, for example, a vision-language model.
 - **construct your own tiny dataset**,
 - e.g., Cultural Assumptions, whether "in the absence of textual cues towards a particular location, models might default to cultures or countries that are better-represented in their training data."
 - see Kolisko, S., & Anderson, C. J. (2024). Exploring Social Biases of Large Language Models in a College Artificial Intelligence Course. *Proceedings of the AAAI Conference on Artificial Intelligence*, 37(13), 15825-15833. <https://doi.org/10.1609/aaai.v37i13.26879>, where Table 1 provides a good example.
 - or, using a dataset from <https://github.com/i-gallegos/Fair-LLM-Benchmark>
 - apply a **potential mitigation strategy** if possible, so that the model is less biased based on your evaluation.
- Prototyping of your project, discussion, and Q&A

Workshop 11 & 18 Feb 2026 – Graph RAG

- Graph Retrieval-Augmented Generation implementation
 - Jupyter notebook provided on ELE
- Prototyping of your project, discussion, and Q&A

13 Feb 2026 – discussion regarding LLM mechanistic interpretability

- Discussion:
 - [Main] Trenton Bricken et al. **Towards Monosemanticity: Decomposing Language Models With Dictionary Learning**. Anthropic, Transformer Circuits Thread, October 2023. <https://transformer-circuits.pub/2023/monosemantic-features/index.html>
 - [Complementary] Adly Templeton. **Scaling Monosemanticity: Extracting Interpretable Features from Claude 3 Sonnet**. Anthropic, Transformer Circuits Thread, May 2024. <https://transformer-circuits.pub/2024/scaling-monosemanticity/index.html>

Guest speaker week on Domain-specific Applications – **GenAI for climate communication**

- Tuesday 17 Feb 2026 **Arjun Biswas**, collaboration with the Met Office

<https://experts.exeter.ac.uk/42763-arjun-biswas>

Topic: RAG, chatbot, AI agents, tools, and human-AI collaboration



Read:

- Arjun Biswas et al. 2025. Evaluating Retrieval Augmented Generation to Communicate UK Climate Change Information. <https://aclanthology.org/2025.climatenlp-1.9/>

Prepare questions if you have

Guest speaker week on Domain-specific Applications – **GenAI for game dev**

- Friday 20 Feb, **Yuqian Sun**, Midjourney, London
<https://fakecheese.me/>, X @cheesetalk1997

Title: Across the Magic Circle: AI Agents as a Creative Medium for Interactive Storytelling



Read:

- Language as Reality: A Co-Creative Storytelling Game Experience in 1001 Nights Using Generative AI
<https://ojs.aaai.org/index.php/AIIDE/article/view/27539>
- AI Nüshu: An Exploration of Language Emergence in Sisterhood Through the Lens of Computational Linguistics.
<https://dl.acm.org/doi/10.1145/3610591.3616427>

Prepare questions if you have